



Review

Opportunities for graphene, single-walled and multi-walled carbon nanotube applications in agriculture: A review



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ABSTRACT

Nanotechnology is an important driver leading to the revolution in the fields of agriculture. Here, we presented insights into the application of carbon nanomaterials (single- and multi-walled carbon nanotubes and graphene) in biomolecule delivery for crop breeding. We also reviewed the application of graphene to biosensors and the utilization efficiency of water, fertilizers and insecticides in agriculture to better manage environmental stresses. Finally, potential molecular interaction mechanisms between graphene and plant roots were discussed. It can be expected that carbon nanomaterials will hold strong promise in developing a more efficient agricultural production system.

1. Introduction

An efficient and sustainable agricultural production system is vital to human survival [1]. Global annual food production requires million tons of fertilizers, pesticides and freshwater [2]. However, agricultural production is facing serious environmental degradation [3,4]. To meet the human demand for food, many new scientific and technological revolutions have emerged, such as new methods (gene editing) and new materials (nanomaterials) [5]. Nanomaterials, normally less than 50 nm, can pass through the biological barriers of plants (such as the cell wall) into the vascular tissue, thus providing a new path for delivering biomolecules, nutrients and pesticides into plants [5].

Nanomaterials have been widely used in electronics, machinery, energy and biomedicine [6–9]. In the field of plants, researchers found that nanomaterials could be absorbed by plants, transported to different tissues and organs, and further improve the seed germination rate and

promote plant growth [10,11]. Nanomaterials can also be used to make sensors for collecting real data on physiological status in plants to enhance the ability of plants in response to environmental stresses to further increase the production of crops and forests [12]. Nanomaterials mainly include metals, metal oxides, polymers, and carbon nanoparticles, of which carbon nanomaterials have received great attention from researchers due to their unique chemical properties, structures and low toxicity [11].

At present, carbon nanomaterials identified mainly include graphene, single-walled carbon nanotubes, multi-walled carbon nanotubes, carbon nanohorns, fullerenes, carbon nano-onions, and carbon nanoquantum dots [13,14]. They can penetrate plant cells to deliver DNA vectors and affect seed germination and plant growth [10,11,15–17]. Graphene is a 2D one-atom thick layer of carbon wrapped into a cylinder to form a single-walled carbon nanotube [18]. Multiple graphene sheets can be made of multi-walled carbon nanotubes wrapped into cylinders inside

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other cylinders of carbon [18,19]. They all have large specific surface areas and can be designed to contain a variety of oxygen-containing functional groups, such as carboxyl groups [20]. In recent years, carbon nanomaterials in agriculture have attracted increasing attention. This paper summarizes the progress of single-walled carbon nanotubes, multi-walled carbon nanotubes, and graphene in crop breeding. In addition, the effects of graphene on plant growth, water, fertilizer and insecticide utilization, environmental stress, the plant-graphene interaction mechanism and its environmental risks were discussed. This review provides a reference for the applications of major carbon

nanomaterials in agriculture.

2. Gene delivery mediated by carbon nanomaterials

Genetic engineering of crops aims to generate pathogen-resistant and high-yielding crops. However, the cell wall in plants is a barrier to exogenous biomolecule delivery. Conventional gene delivery methods, including *Agrobacterium*-mediated genetic transformation and gene guns, are inefficient, damaging the plant tissue, and are only successful in a few plant species and genotypes [21,22]. Compared to the 300–500 nm

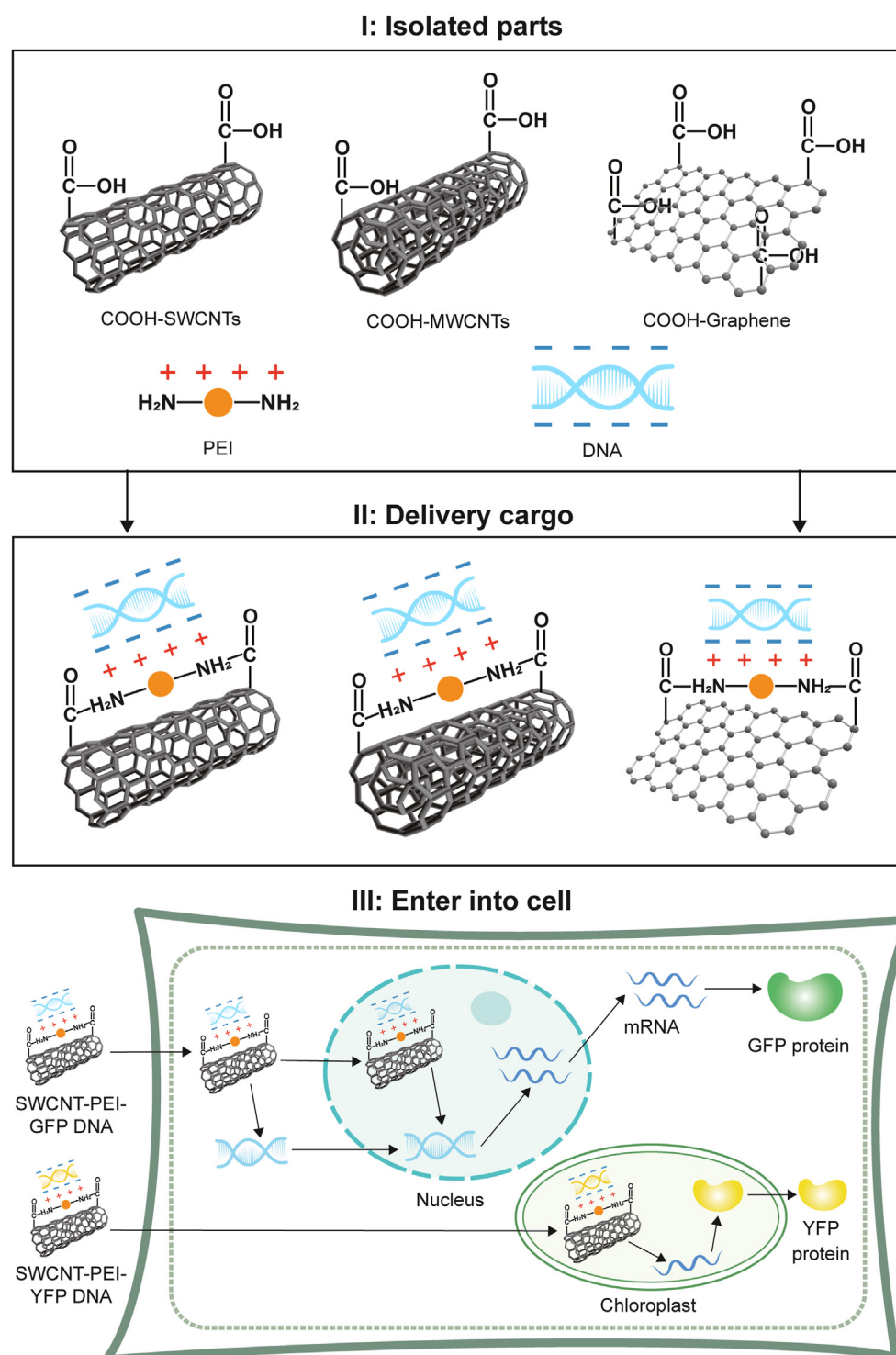


Fig. 1. Delivery cargo based on the carboxylated carbon nanomaterial-PEI-DNA complex by electrostatic attraction and subsequent entry into plant cells for gene expression. First, carboxylated SWCNTs, MWCNTs and graphene were covalently modified with poly-ethylenimine (PEI) to carry a net positive charge. Next, negatively charged phosphate groups on DNA was grafted onto positively charged PEI-COOH-SWCNTs, MWCNTs or graphene cargo by electrostatic attraction. Notes: SWCNTs: Single-walled carbon nanotubes, MWCNTs: multi-walled carbon nanotubes, PEI: poly-ethylenimine, GFP: green fluorescent protein, YFP: yellow fluorescent protein.

size exclusion limit of the animal cell membrane, the plant cell wall excludes particles with sizes greater than approximately 50 nm [22]. Nanoparticles, with a size of at least under 50 nm, are promising materials for delivering biomolecules into plant cells for genetic engineering to produce crops with desirable traits.

2.1. Gene delivery mediated by single-walled carbon nanotubes

Single-walled carbon nanotubes (SWCNTs) generally have a 1 nm diameter and easily cross the barrier of the plant cell wall [23]. Because of this property, researchers have developed a method to graft DNA onto SWCNTs by electrostatic attraction. First, carboxylated single-walled carbon nanotubes (COOH-SWCNTs) were covalently modified with poly-ethylenimine (PEI) to carry a net positive charge. Next, positively charged PEI-COOH-SWCNTs are incubated with negatively charged phosphate groups on plasmid DNA vectors (Fig. 1). A recent study used this cargo tool to achieve highly efficient green fluorescent protein (GFP) gene delivery and transient expression without transgene integration in both dicot and monocot plant mature leaf and protoplast cells (Fig. 1), including *Nicotiana benthamiana*, *Eruca sativa*, *Gossypium hirsutum*, and *Triticum aestivum* [24]. Based on this strategy and principle, researchers from the same group designed chitosan-complexed single-walled carbon nanotubes to selectively deliver the yellow fluorescent protein (YFP) gene loaded on a plasmid vector to chloroplasts of different plant species, including mature leaf cells in *Eruca sativa*, *Nasturtium officinale*, *Nicotiana tabacum* and *Spinacia oleracea*, as well as mesophyll protoplasts in *Arabidopsis thaliana*. This chloroplast-targeted YFP gene delivery also showed a transient expression mode [25]. A subsequent study used DNA nanostructures to deliver siRNAs and effectively silence a constitutively expressed GFP gene in *Nicotiana benthamiana* leaves. siRNA-mediated gene silencing delivered by DNA nanostructures led to the most considerable GFP silencing effect on Day 3 but returned to normal expression levels by Day 7, which presented a transient expression phenomenon, like a flash in the pan [26].

2.2. Gene delivery mediated by multi-walled carbon nanotubes

Unlike SWCNTs, multi-walled carbon nanotubes (MWCNTs) reduce their efficiency in delivering DNA into plant cells due to their larger particle size [24]. Indeed, the MWCNTs nanocarriers demonstrated the ability of transformation in *Nicotiana tabacum* protoplasts (Fig. 1) [27]. Other studies also indicated that an endosome-escaping uptake mode of MWCNTs penetrates the cell membrane of *Catharanthus roseus* protoplasts [28]. These findings demonstrated the advantages of multi-walled carbon nanotubes in plant gene delivery.

2.3. Gene delivery mediated by graphene

Graphene has been successfully selected as a nanocarrier to transport genes into animal cells. For instance, a graphene delivery system was used for the transfection of pEGFP-p53 into breast cancer cell lines [29]. Subsequent studies showed that modified graphene quantum dots could also be used to deliver intact and functional mRNA to human Huh-7 hepatocarcinoma cells [30]. Surface-functionalized graphene oxide nanoflakes were applied for gene delivery in human cervical cancer cells and embryonic kidney cells [31]. Graphene oxide and poly-ethylenimine (PEI) conjugates showed a high enhanced green fluorescent protein (EGFP) transfection efficiency in live cells [32]. Graphene oxide substrates served as a valid platform to preconcentrate PEI/pDNA complexes and exhibited successful gene delivery and gene transfection in various cell lines, including stem cell research and tissue engineering [33].

At present, graphene nanomaterials have been less studied for gene delivery into plant cells. However, carbon nanotubes (CNTs) have been reported to successfully deliver DNA into many non-model plant cells and produce a high expression of genes without integration into the plant nuclear genome (Fig. 1) [34,35]. Graphene-mediated transformation

technologies in agriculture have the potential to change crop genetic modification by incorporating foreign vectors of DNA, mRNA, RNAi or CRISPR/Cas9 into the crop host genome in the future [36]. Although great progress has been made in carbon nanomaterial-mediated delivery approaches, the transformed plants obtained are all transiently expressed, and stable transformation plants cannot be obtained, which is the direction of breakthrough in the future.

3. Opportunities for graphene in the agriculture technology revolution

In addition to its application in gene delivery, graphene also affects the growth and development of plants [13,37,38]. Current studies on the effect of graphene on plants are mainly focused on the following directions: graphene biosensors [38,39], effects of graphene on plant physiology [40], seed germination [40–42], seedling root growth [42], plant biomass and food quality as well as utilization efficiency of water, fertilizers and pesticides in agriculture [37,43] (Fig. 2).

3.1. Graphene biosensors

The determination method for genotype traits of crops, such as PCR, ELISA and DNA hybridization, either are labor intensive or require bulky and expensive instruments. Biosensors have enormous advantages of easy operation, short turnaround time, high accuracy, and wide detection capacity [44]. Graphene has been used for fluorescence biosensors [45], biomolecular labels [46], enzymatic biosensing [47], DNA sensing [39], and immunosensing [48] (Fig. 2). In addition, Berry's group first reported a graphene-based biosensor for achieving single bacterial detection [49]. Chen et al. demonstrated that graphene-based biosensors can detect *E. coli* bacteria [50]. A recent breakthrough shows that the interaction between biomaterials and graphene can be employed as a nanosensor platform for monitoring bacterial growth [51]. These results indicate that graphene might be used as a biosensor to quickly sense and obtain target desirable trait information in crops, which would be helpful to accelerate the process of breeding selection.

3.2. Improvement of graphene in the utilization efficiency of water, fertilizers and pesticides in agriculture

Graphene can prevent soil water from evaporating due to its hydrophilic oxygen-containing functional groups [52]. Therefore, graphene can enhance the soil and water conservation capacity of plants and improve water use efficiency. Because of this, graphene can promote seed germination by increasing the moisture content of seeds (Fig. 2). For instance, studies have shown that graphene can promote the germination of tomato seeds [37] and spinach seeds [41].

Plants need an external environment to provide macroelements, such as N, P, and K, and microelements, such as Mn, Mo, Zn, Cu, Fe and B, for their life activities. Excessive or low fertilization efficiency will lead to eutrophication of soil and water and pollute the environment [53]. Graphene can efficiently deliver nutrients through roots, leaves and seed coat agents to improve nutrient utilization efficiency [43,54,55]. For example, Zhang et al. [43] exhibited slow-release KNO₃ fertilizer encapsulated by graphene films with the process of releasing fertilizer extended to 8 h in water (Fig. 2). Kabiri et al. [54] developed a graphene-based carrier that can load the plant micronutrients Zn and Cu with controllable slow release. They also demonstrated that Zn and Cu uptake by wheat was higher when using graphene-based carriers than the control. Li et al. [55] created graphene-based controlled-release fertilizers with nutrient release lasting for approximately a week. In view of this, industrial-scale production of graphene-based fertilizers will become feasible over time.

Benefitting from the efficient utilization of water and fertilizer in the soil, graphene can promote the growth and development of aboveground plant parts. For instance, a study showed that graphene can elevate the

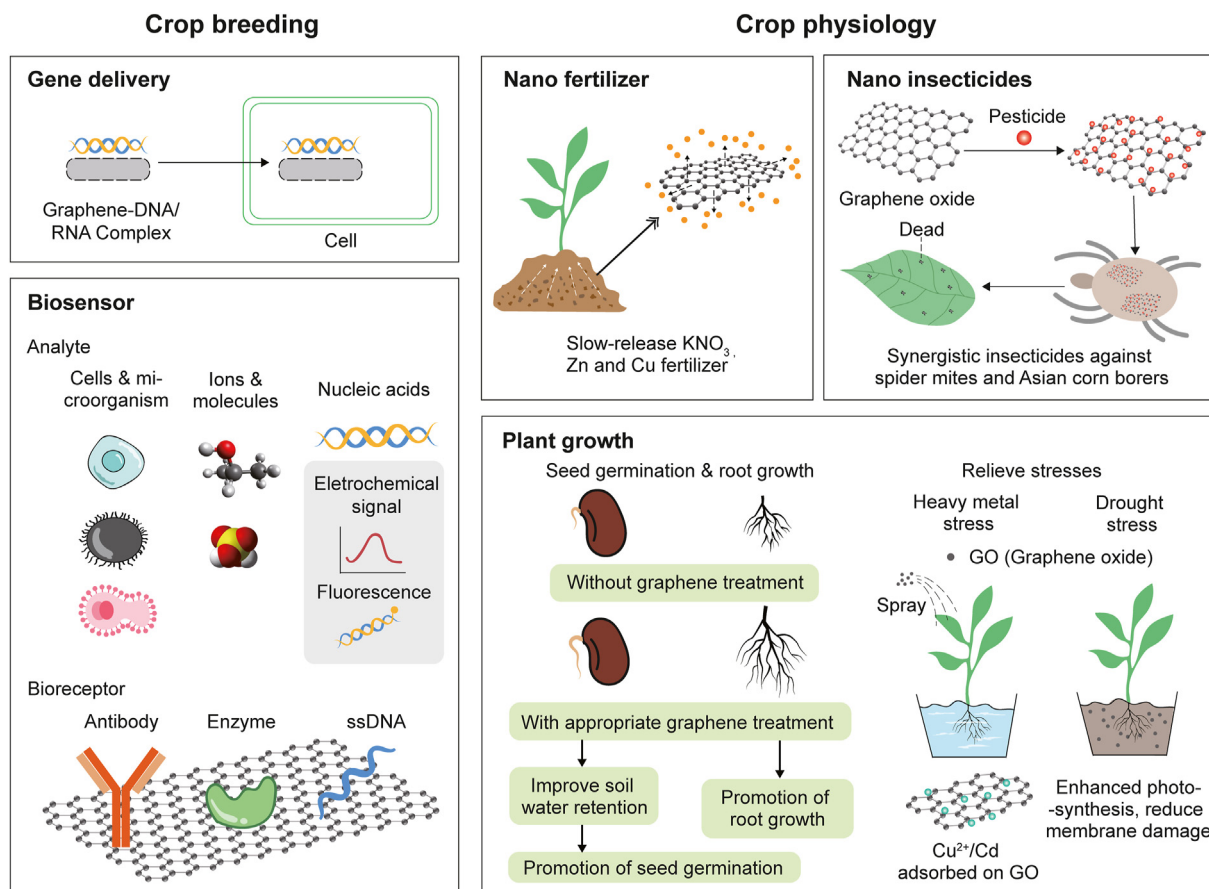


Fig. 2. Current applications of graphene in agriculture. Graphene applications in agriculture include two aspects of crop breeding and crop physiology. Crop breeding was mainly involved in the graphene-based gene delivery and graphene biosensors. Crop physiology was focused on the effects of graphene on plant growth, such as seed germination and root growth, as well as utilization efficiency of water, fertilizers and pesticides in agriculture.

water content of broad bean seeds and improve the growth status of broad bean plants [56]. Graphene also significantly increased the stem and leaf fresh weight in rice [42] and promoted the growth of tomato seedlings by regulating gibberellic acid synthesis [37]. Importantly, graphene greatly promoted the growth of roots, stems, leaves, flowers and fruits of coriander and garlic plants, finally promoting their yields [57].

Moreover, graphene as a pesticide delivery vector could be used for pest management. Wang et al. [58] reported that graphene can serve as a carrier of pesticides for enhancing acaricidal activity against spider mites (Fig. 2). A continuous study showed that graphene also had multifunctional synergistic activity toward insecticides against Asian corn borers [58,59]. Graphene can be a carrier of pesticides to be adsorbed on the surface of insects and disrupt the cement layer cell of insects to improve the dispersibility and utilization efficiency of insecticides. A recent study found that graphene can make the Asian corn borer “fatter” but shorten its life span [60]. This kind of graphene-based pesticide delivery system would have wide application for plant protection in the future.

3.3. Promotion of graphene in plant management of drought, pathogen and heavy metal stressors

Biological and abiotic stresses can reduce the yield and quality of agricultural plants. Biological stresses include pathogens and weeds, and abiotic stresses include high temperature, freezing injury, drought, flooding and salinity. These environmental stress factors can cause more than 50% loss of crop yield [61,62]. Additionally, many abiotic stresses make plants more susceptible to pathogen infection and increase field loss [63].

The effects of graphene on plant growth under stress conditions attract more concern. Graphene can relieve drought stress of *Paeonia ostii* as an effective soil water retention agent due to its hydrophilic oxygen-containing functional groups [52]. Under drought stress, graphene treatment enhanced photosynthesis, increased the number of intact mesophyll cells and organelles, opened stomata, and increased antioxidant enzyme activities in *P. ostii* plants [52] (Fig. 2). Other studies also indicated that graphene treatment increased photosynthesis in *Vicia faba* [64] and lettuce [65] plants. The significantly improved photosynthetic efficiency in plants would help to relieve environmental stresses.

Meanwhile, graphene-based materials showed remarkable bactericidal activity in plants [66]. He et al. found that graphene could prolong the vase life of cut flowers as an antibacterial agent [67]. Wang et al. [68] showed that graphene could inhibit the growth of *Fusarium graminearum* in wheat and revealed the molecular mechanism of its inhibition from the levels of transcriptomics, proteomics and metabolomics, indicating the role of graphene in plant protection.

In addition, studies have shown that graphene can help plants protect themselves from heavy metal damage. Foliar application of graphene would improve biomass and reduce the oxidative damage of Cd-stressed lettuce [61] and decrease Cd toxicity on lettuce [69] (Fig. 2). A study also proved that graphene could alleviate Cu stress in *Lemna minor* [70] (Fig. 2). At present, there are few reports on the effects of graphene on plant resistance under external stresses, and the involved mechanisms need to be further studied. Based on the analysis above, graphene has broad prospects for improving the resilience of agricultural systems and making plants better able to manage environmental changes. The in-depth studies of graphene's effects on plant resistance under environmental stresses will provide a new nanobiotechnology approach for

the improvement of saline-alkali lands.

3.4. Graphene mediated plant root growth in a dose-dependent manner and their interaction mechanisms

High concentrations of graphene inhibit the growth and development of plants [71]. For instance, 100 mg/L graphene inhibited the growth and development of *Vicia faba* plants [64]. Begum et al. [40] reported that 1000 mg/L graphene led to a large accumulation of reactive oxygen species (ROS) in the roots of tomatoes, red spinach and cabbage and eventually inhibited the growth of these plants. Liu et al. [42] reported that 100 mg/L graphene repressed the growth and development of rice roots and reduced plant biomass. These results indicated that high concentrations of graphene may cause strong osmotic stress in plant cells, causing them to lose water to death.

Treating plants with low doses of graphene nanoparticles improves their root growth and development. Studies have shown that 2–5 mg/L graphene can promote the root growth of *Rubus corchorifolius* [72,73], *Chenopodium quinoa* [74], and rice plants [42]. Graphene (50 mg/L) markedly increased the total root length and root volume of maize plants [75], improved the morphological characteristics of roots in *Aloe vera* [76] and significantly increased the root tips and hairs in tomato plants [77].

Transcriptome research revealed the upregulated expression of transcription factors, plant hormone signal transduction, nitrogen and potassium metabolism, and secondary metabolism genes in maize roots in response to graphene treatment [75]. Another study found that proline synthesis genes were induced by graphene, but gene responses to biotic stimuli were inhibited in *Pinus tabuliformis* roots [78]. In tomato roots, graphene significantly increased the root auxin content and induced the

expression of root development-related genes (*SlExt1* and *LeCTR1*) [77].

A recent study used graphene to treat 48 plant species to investigate its effects on plant root growth [79]. The results showed a promotion effect in 65% of angiosperms, inhibition in 22% of angiosperms, and no effect on 13% of angiosperms and two gymnosperms. Differentially expressed genes (DEGs) from RNA-seq data were mainly enriched in pathways including starch and sucrose metabolism, glycolysis/gluconeogenesis, pyruvate metabolism, citrate cycle (TCA cycle), phenylpropanoid biosynthesis, glutathione metabolism, endocytosis, peroxisome, etc. For the plant roots showing promotion effects, graphene increased the activities of antioxidant enzymes and induced the gene expression of the respiration pathways, leading to elevated mitochondrial respiration function with increasing plant roots. In contrast, the repressed expression of genes in these pathways inhibited plant roots and reduced antioxidant enzyme activities as well as weakened respiration, leading to a decrease in root length [79]. In addition, graphene also induced differentially expressed genes related to lignin biosynthesis, photosynthesis-antenna proteins, carbon fixation in photosynthetic organisms, and glyoxylate and dicarboxylate metabolism [52].

Further experimental data showed that 25 mg/L graphene elevated the contents of ammonium ions (NH_4^+) and potassium (K^+) in the soil around the root in *Z. mays* and *V. faba* plants [64,75]. Early studies also indicated that graphene had an obvious retention effect on total nitrogen, phosphorus, and potassium in the soil [80,81]. These results confirmed that graphene had a significant retention effect on the nutrients after being watered into the soil and improved the soil fertility to promote the growth of plant roots. Graphene is mainly composed of C atoms, and its surface is enriched in oxygen-containing functional groups, such as carboxyl ($-\text{COOH}$) and hydroxyl ($-\text{OH}$) groups. These functional groups and cation- π interactions [82] in graphene at low concentrations can

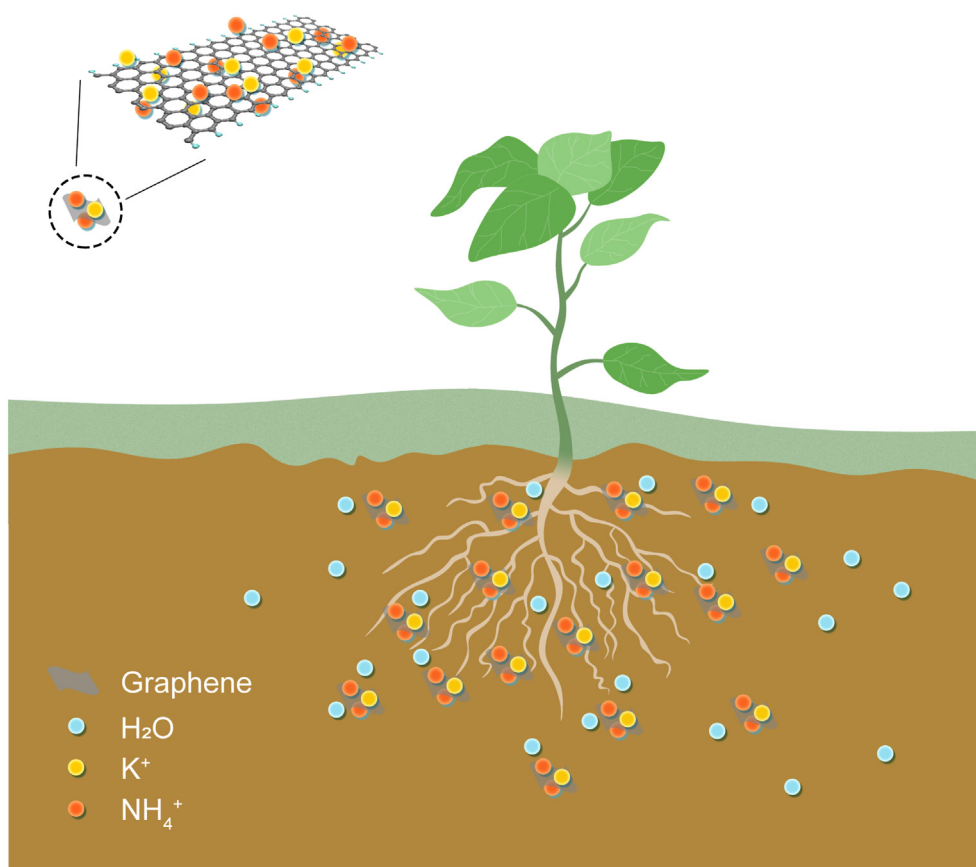


Fig. 3. The interaction model of graphene and plant roots. A model for the mechanism underlying the graphene effect on the root growth of plants. When the roots of plants are exposed to graphene, the H_2O , NH_4^+ and K^+ contents of the soil are elevated because of hydrogen bonding, electrostatic attraction and cation- π interactions, which lead to the accumulation of H_2O , NH_4^+ and K^+ on the graphene surface [82].

attract cations (e.g., NH_4^+ and K^+) in the soil, providing nutrient elements for plants to promote root development (Fig. 3).

3.5. Risks of graphene on the environment

The use of graphene in food crops would cause potential safety concerns for sensitive consumers [83]. Researchers used ^{14}C -labeled graphene to determine its uptake, distribution, transformation, and depuration in rice and found that graphene can be quickly transported from roots to shoots and leaves [84]. ^{14}C -labeled graphene can penetrate the cell wall and membrane into the chloroplast of leaves and react with OH^- in leaves, contributing to the degradation of graphene by mineralization into $^{14}\text{CO}_2$. Long-cycle experiments showed that although rice was able to ingest graphene, the accumulation of graphene in stems and leaves disappeared at 15 days after exposure to graphene, and no graphene remained in harvested rice [84]. In addition, it was found that graphene can be degraded by microorganisms in the soil environment. Scholars have confirmed that the polycyclic structure of graphene is similar to that of lignin and polycyclic aromatic hydrocarbons, so graphene can be broken down and degraded by lignin enzymes secreted by some fungi [85]. Qu et al. screened a bacterium from soil that was able to utilize graphene as the sole carbon source to support growth [86]. These studies would be helpful to eliminate the public perception of safety concerns surrounding the use of engineered graphene in agricultural production.

4. Future directions in agriculture

Sustainable, efficient and resilient agriculture systems would benefit from carbon nanomaterial-based technologies. Applications of carbon nanomaterials in agriculture are emerging and will continue to advance the fundamental understanding of carbon nanomaterial-plant interactions. Another direction of carbon nanomaterial application in agriculture is improving the efficiency of biomolecule and agrochemical delivery, including delivery time, location, dose and form. Therefore, it can be anticipated that carbon nanomaterials, including single-walled carbon nanotubes, multi-walled carbon nanotubes and graphene, may become feasible to produce 'smart plants' in the future.

Author contributions

ZWC and JGZ conceived the project; YYZ, ZWC and SJ drew the diagrams. ZWC, JFC, YYZ, JQH, ZSZ, JWL, SJ, JQ, BYX and JZ wrote and revised the manuscript. All authors have read and approved the manuscript.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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